

Exp No: 11

STUDY OF GPIO & BIT ADDRESSING IN AT89C51

PROGRAM:

```
#include <reg51.h>
```

```
sbit LED1 = P1^0;
```

```
sbit LED2 = P1^1;
```

```
void delay()
```

```
{
```

```
    unsigned int i,j;
```

```
    for(i=0;i<200;i++)
```

```
        for(j=0;j<1000;j++);
```

```
}
```

```
void main()
```

```
{
```

```
    while(1)
```

```
    {
```

```
        LED1 = 1;
```

```
        LED2 = 0;
```

```
        delay();
```

```
        LED1 = 0;
```

```
        LED2 = 1;
```

```
        delay();
```

```
    }
```

```
}
```

OUTPUT:

```
10: void main()
11: {
12: while(1)
13: {
```

exp11.c STARTUP.A51

```
1  #include <reg51.h>
2  sbit LED1 = P1^0;
3  sbit LED2 = P1^1;
4  void delay()
5  {
6      unsigned int i,j;
7      for(i=0;i<200;i++)
8      for(j=0;j<1000;j++);
9  }
10 void main()
11 {
12 while(1)
13 {
14     LED1 = 1;
15     LED2 = 0;
16     delay();
17     LED1 = 0;
18     LED2 = 1;
19     delay();
20 }
21 }
```

Parallel Port 1

Port 1

P1: 0xFE 7 Bits 0

Pins: 0xFE ✓✓✓✓✓✓✓